

SIMATS ENGINEERING
MACHINE LEARNING (ITA0601)
STUDENT ACADEMICS
FACULTY: DR.R.KAVITHA
NAME: HARSHAVARDHAN
REG NO: 192324189

QUESTION 1

Design a pseudocode algorithm to read a dataset from a file and preprocess it using array and DataFrame operations. The algorithm should handle missing values, remove duplicate records, normalize numerical columns, and standardize categorical values. Include steps for indexing, filtering invalid entries, and saving the cleaned dataset. Ensure the algorithm clearly defines input, processing steps, and output while maintaining computational efficiency.

SOLUTION

Pseudocode Algorithm:

```
BEGIN
INPUT: dataset file path and valid constraints/ranges
Read the file into a Pandas DataFrame.
Inspect columns, data types, missing values, and duplicate
records. Use indexing to select required columns and rows.
Filter invalid entries using Boolean conditions and predefined valid ranges.
Handle missing numerical values using an appropriate statistic such as median.
Handle missing categorical values using mode or a defined valid category.
Remove duplicate records using duplicate detection.
Identify numerical columns and normalize them, for example using
Min-Max scaling.
Standardize categorical values by trimming spaces, changing case, and mapping
equivalent labels.
Validate that required fields contain no missing values and numerical values
satisfy constraints.
Save the cleaned DataFrame to a new output file.
OUTPUT: cleaned and standardized dataset.
END
```

This approach separates input, processing, validation, and output stages so that the required operations are clear and computationally efficient.

QUESTION 2

Write a detailed pseudocode algorithm to compute statistical measures such as mean, median, standard deviation, and variance from a large dataset. The algorithm should demonstrate how to efficiently use array-based operations, apply filtering to exclude unwanted values, and perform aggregation. Include steps for validating the dataset and handling edge cases such as empty arrays or extreme outliers.

SOLUTION

Pseudocode Algorithm:

```
BEGIN
INPUT: large numerical dataset
Load the numerical values into a NumPy array.
Validate that the array exists, has numeric values, and is not empty.
Use indexing/filtering to remove missing, invalid, or unwanted values.
Detect extreme outliers using an appropriate rule such as Z-score or
IQR. Decide whether to exclude or cap extreme values according to the
analysis requirement.
Use vectorized NumPy aggregation functions to calculate mean, median, variance,
and standard deviation.
For a very large dataset, process values in chunks to reduce memory
usage. Check the number of valid observations before calculating each
statistic. If the valid array is empty, report that statistics cannot be
calculated.
Validate results by checking ranges, data types, and a small sample manually.
OUTPUT: mean, median, variance, and standard deviation with validation
information.
END
```

This approach separates input, processing, validation, and output stages so that the required operations are clear and computationally efficient.

QUESTION 2

Develop a pseudocode algorithm to combine two or more DataFrames containing related data. The algorithm must include steps for aligning indices, merging datasets based on a common key, removing duplicates, and validating the integrity of the merged data. Incorporate grouping and sorting operations, and ensure that constraints such as consistency and completeness of records are maintained.

SOLUTION

Pseudocode Algorithm:

```
BEGIN
INPUT: two or more related DataFrames and a common key.
Load and inspect all DataFrames.
Check the common key for missing values and duplicate records.
Standardize the key data type and format across all DataFrames.
Align indices or reset indices where necessary.
Merge DataFrames using the common key and the required join type.
Use grouping by the common key to detect unexpected duplicate records.
Remove only confirmed duplicate records while preserving valid entries.
Use filtering to identify unmatched or incomplete records.
Sort the merged DataFrame by the common key.
Set or reset the final index to maintain consistent indexing.
Validate record counts, unique key counts, missing values, and required columns.
OUTPUT: complete, consistent, sorted, and validated merged DataFrame.
END
```

This approach separates input, processing, validation, and output stages so that the required operations are clear and computationally efficient.

QUESTION 2

Create a pseudocode algorithm that filters a dataset based on multiple conditions (e.g., value ranges, categories) and then sorts the filtered results. The algorithm should utilize indexing techniques, logical conditions, and efficient sorting methods. Include steps for handling large datasets and optimizing performance while ensuring that the output meets all filtering constraints.

SOLUTION

Pseudocode Algorithm:

```
BEGIN
INPUT: dataset, filtering conditions, and sorting column(s)
Load the dataset into a Pandas DataFrame or suitable array structure.
Identify the required columns using indexing.
Define all filtering conditions, such as numerical ranges and allowed
categories.
Combine conditions using logical AND/OR operators.
Apply Boolean indexing to select only valid records.
Validate that every remaining record satisfies all required conditions.
For large datasets, process data in chunks when necessary and avoid unnecessary
copies.
Sort the filtered result using an efficient sort_values operation.
Select only the required output columns.
Validate the sorted order and record count.
Save or return the filtered and sorted
dataset.
OUTPUT: dataset satisfying all filtering conditions in the required
sorted order.
END
```

This approach separates input, processing, validation, and output stages so that the required operations are clear and computationally efficient.

QUESTION 2

Write a pseudocode algorithm for a complete data analysis workflow that includes reading data from files, converting it into DataFrames, performing grouping and aggregation, and generating plots. The algorithm should specify steps for handling file input/output errors, applying transformations, and producing summarized results suitable for visualization. Ensure clarity in each stage from data loading to final output generation.

SOLUTION

Pseudocode Algorithm:

```
BEGIN
INPUT: one or more data files and user-defined analysis conditions
Identify the input files.
Attempt to read each file using the appropriate Pandas file-reading
function. If a file cannot be opened or has an invalid format, record the
error and continue with valid files where appropriate.
Convert the loaded data into Pandas DataFrames.
Inspect column names, data types, missing values, and inconsistent formats.
Apply required transformations such as type conversion, missing-value handling,
and standardization.
Filter records using user-defined conditions.
Group the processed DataFrame by the required categorical or key
columns. Apply aggregation functions such as sum(), mean(), count(), or
median(). Sort the summarized results when required.
Generate suitable plots such as bar, line, histogram, or scatter plots from the
summarized data.
Add meaningful titles, axis labels, and legends where required.
Save the summarized results and visualization outputs.
OUTPUT: validated DataFrame, summarized report, and meaningful plots.
END
```

This approach separates input, processing, validation, and output stages so that the required operations are clear and computationally efficient.